Lecture 16: Confidence Intervals for Proportions

Motivating example: A drug company is testing "LDL-Be-Gone", a newly-developed cholesterol medication. The obvious question: Does it work?

STEP 1: Conduct a well-controlled experiment on 100 volunteers, noting that only 10 of the 50 not using the drug saw a significant decrease in their bad cholesterol levels, while 32 of the 50 using the drug saw a significant decrease.

STEP 2: Spend $1B mass-producing and marketing the drug?

Recall: Suppose we randomly sample n members of a large population and classify proportion $\hat{p}$ of them as "successes". If p represents the proportion of "successes" in the population at large, the sampling distribution of $\hat{p}$ is ...

Then by the 68-95-99.7 rule, approximately 95% of sample proportions $\hat{p}$ will fall between ...

Getting more precision with a z-table or Excel:

Definitions: **critical value**, **margin of error**, **confidence interval**

Approximating standard error:

Example: We are interested in the proportion p of students at UNC who have engaged in binge drinking behavior at least once within the last month. In a SRS of 485 UNC students, we find that 185 have done so.

- Find an (approximate) 95% confidence interval for p.

- Now suppose your sample size is 4*485=1940 and you have the exact same proportion of binge drinkers. Construct another 95% confidence interval based on this larger sample.

Interpretation of "95% confidence"

Other confidence levels

Example: Find z^* for the following confidence levels:

- 90%

- 99%

Example: The probability of rolling a seven with a pair of fair dice is $\frac{1}{6}$. We have a pair of dice that may or may not be fair; let p be the unknown probability they will produce a seven.

Suppose we roll this pair of dice 120 times and observe a seven 25 times.

- Construct a 95% confidence interval for p based on these results.

- Construct a 99% confidence interval for p based on these results.

Shrinking margin of error:

Choosing a sample size to get desired margin of error

Example: We plan to take a poll to estimate p, the proportion of American voters who support throwing out the current health care system and putting Canada ("America's hat") in charge of US health care. We want our 95% margin of error to be ≤ 0.03.

- If we think around 25% of Americans feel this way, how big of a sample do we need?

- What if we have no idea about the value of p?